

A Reliable Multi-Hop Multicast Protocol for Dynamic Multi-Rate Wireless Systems

Project report submitted in partial fulfillment
of the requirements for the degree of

Bachelor of Technology
in
Communication and Computer Engineering

by

Nilesh Goyal - Roll No. 23UCC624
Mudit Choudhary - Roll No. 23UCC623
Kushagra Garg - Roll No. 23UCC564

Under Guidance of
Dr. Prateek Rathore



Department of Communication and Computer Engineering
The LNM Institute of Information Technology, Jaipur

April 2026

The LNM Institute of Information Technology
Jaipur, India

CERTIFICATE

This is to certify that the project entitled “A Reliable Multi-Hop Multicast Protocol for Dynamic Multi-Rate Wireless Systems” , submitted by Nilesh Goyal (23UCC624), Mudit Choudhary (23UCC623) and Kushagra Garg (23UCC564) in partial fulfillment of the requirement of degree in Bachelor of Technology (B. Tech), is a bonafide record of work carried out by them at the Department of Communication and Computer Engineering, The LNM Institute of Information Technology, Jaipur, (Rajasthan) India, during the academic session 2026-2027 under my supervision and guidance and the same has not been submitted elsewhere for award of any other degree. In my/our opinion, this report is of standard required for the award of the degree of Bachelor of Technology (B. Tech).

Date

Adviser: Dr. Prateek Rathore

Acknowledgments

We would like to express our sincere appreciation to Dr. Prateek Rathore for his continuous guidance, steadfast support, and valuable feedback throughout our BTech project titled “A Reliable Multi-Hop Multicast Protocol for Dynamic Multi-Rate Wireless Systems.” His expertise and mentorship played a crucial role in transforming our ideas into a well-structured and in-depth study.

His thoughtful suggestions and constructive feedback consistently motivated us to improve our work and push beyond our limits. We are truly grateful for the time, knowledge, and resources he generously shared, which not only strengthened this project but also deepened our understanding of the subject.

Abstract

This project presents a reliable multi-hop multicast protocol designed for dynamic multi-rate wireless systems. In such networks, nodes communicate over multiple hops while experiencing varying transmission rates due to changing channel conditions, interference, and mobility. Traditional multicast protocols often assume fixed link rates and independent packet losses, which limits their effectiveness in real-world scenarios.

To overcome these limitations, the proposed protocol incorporates three key factors: link cost, link rate, and inter-link correlation. Link cost, typically measured using metrics like Expected Transmission Count (ETX), estimates the reliability of packet delivery. Link rate accounts for variations in data transmission speeds across nodes, enabling the protocol to adapt to dynamic bandwidth conditions. Additionally, link correlation is considered to capture the dependency between packet losses on different links, which often arises due to shared environmental factors.

By combining these parameters, the protocol selects an efficient set of forwarder nodes that reduces redundant transmissions while improving delivery performance. The correlation-aware approach leads to more realistic routing decisions compared to traditional methods.

Simulation results under varying network conditions demonstrate improved packet delivery ratio and reduced transmission overhead. Overall, this work highlights the importance of integrating real-world wireless characteristics into multicast protocol design for achieving reliable and efficient communication.

Contents

List of Figures	viii
List of Tables	ix
1 Introduction	1
1.1 The Area of Work	1
1.2 Problem Addressed	1
1.3 Existing System	2
1.3.1 Single-Path Routing	2
1.3.2 Metric-Based Routing (ETX-Based Systems)	2
1.3.3 Opportunistic (Anypath) Routing	3
1.3.4 Limitations of Existing Systems	3
2 Literature Review	4
2.1 Introduction	4
2.2 Multicast Least Cost Anypath Routing (M-LCAR)	4
2.3 MORE: MAC-Independent Opportunistic Routing Encoding	5
2.4 CodePipe: Reliable Multicast with Pipelined Network Coding	5
2.5 Reliable Relay Assisted Wireless Multicast Using Network Coding	5
2.6 Polynomial-Time Algorithms for Multirate Anypath Routing	6
2.7 Pacifier: High-Throughput Reliable Multicast	6
2.8 Research Gap and Motivation	6
3 Proposed Work : Multirate Multicast Routing	7
3.1 The Proposed Solution	7
3.2 Proposed Method in the Context of Routing Algorithms	8
3.3 Determining ETT	8
3.3.1 One Source Two Destination	8
3.3.2 One source Two Intermediate Single Destination	9
3.3.3 One source Two Intermediate Two Destination	11
3.3.4 One source Three Intermediate Two Destination	13
3.3.5 One source Three Intermediate Three Destination	15
4 Packet Simulation and Results	17
4.1 Single Source Single Destination	17
5 Conclusions and Future Work	18
5.1 Conclusion	18

5.2 Future Work	18
---------------------------	----

Bibliography	18
---------------------	-----------

List of Figures

3.1	Single Source Two Destination	8
3.2	Single Source Single Destination Using Two Intermediate	9
3.3	Single Source Two Destination Using Two Intermediate	11
3.4	Single Source Two Destination Using Three Intermediate	13
3.5	Single Source Two Destination Using Three Intermediate	15

List of Tables

3.1 Forwarder Set vs Destination Set	11
--	----

Chapter 1

Introduction

1.1 The Area of Work

Routing in wireless multi-hop networks is challenging due to high packet loss, dynamic link quality, and varying transmission rates. Traditional single-path routing relies on a fixed next-hop, which often leads to packet loss when the chosen link performs poorly, even if other neighboring nodes could have successfully received and forwarded the data. To address this limitation, this project focuses on multicast routing using opportunistic (anypath) principles, where multiple neighboring nodes act as potential forwarders, improving the chances of successful packet delivery in unreliable wireless environments.

The main area of work involves enhancing multicast routing by incorporating link cost, link rate, and inter-link correlation into the forwarder selection process. By considering real-world dependencies between links along with varying transmission rates, the proposed approach enables more efficient and practical routing decisions. Overall, the project aims to develop a correlation-aware, rate-adaptive multicast protocol that improves reliability while minimizing redundant transmissions in dynamic wireless networks.

1.2 Problem Addressed

Wireless multi-hop networks face significant challenges due to high packet loss, fluctuating link quality, and dynamic transmission rates. Traditional routing protocols, especially single-path approaches, are not well-suited for such environments as they rely on a fixed next-hop, making them vulnerable to link failures and poor performance. Even opportunistic routing techniques often assume that links are independent and operate at uniform rates, which is not true in real-world wireless conditions.

This project addresses these limitations by considering link cost, link rate, and inter-link correlation in multicast routing. It focuses on overcoming the inefficiencies caused by ignoring

link dependencies and rate variations, which can lead to redundant transmissions and reduced reliability. The goal is to design a more realistic and efficient routing strategy that improves packet delivery while adapting to dynamic network conditions.

1.3 Existing System

Existing routing protocols in wireless multi-hop networks primarily rely on single-path routing, where each node forwards packets through a fixed next-hop. These approaches use metrics such as hop count or link cost (e.g., ETX) to determine the best path. However, they often perform poorly in dynamic wireless environments, as they do not adapt well to frequent changes in link quality and may result in high packet loss when the selected path degrades.

More advanced approaches like opportunistic (anypath) routing improve performance by allowing multiple potential forwarders, increasing the chances of successful packet delivery. However, most existing systems assume that links are independent and operate at fixed transmission rates. This assumption ignores real-world factors such as inter-link correlation and varying link rates, leading to inefficient forwarder selection, redundant transmissions, and sub-optimal network performance.

1.3.1 Single-Path Routing

Traditional routing protocols in wireless multi-hop networks follow a single-path approach, where each node forwards packets through a predetermined next-hop toward the destination. The path is selected using simple metrics such as hop count, delay, or link cost (e.g., ETX). While this method is straightforward and easy to implement, it is not well-suited for wireless environments where link conditions are highly dynamic. A path that appears optimal at one moment may quickly degrade due to interference, mobility, or environmental changes. As a result, packets are frequently lost or require retransmissions, increasing delay and reducing overall network efficiency. Moreover, this approach fails to exploit the broadcast nature of wireless communication, where multiple neighboring nodes may overhear the same transmission.

1.3.2 Metric-Based Routing (ETX-Based Systems)

To improve routing performance, several protocols use advanced link-quality metrics such as Expected Transmission Count (ETX), which estimates the average number of transmissions required for successful packet delivery over a link. ETX-based routing selects paths with higher reliability, reducing retransmissions compared to simple hop-count methods. These systems provide better performance in moderately dynamic environments by accounting for packet loss probabilities. However, despite these improvements, they still rely on selecting

a single “best” path. This limits their effectiveness in highly dynamic conditions, as they do not take advantage of alternative paths that may be available at any given time. Additionally, ETX does not inherently consider link rate variations, which can lead to inefficient bandwidth utilization.

1.3.3 Opportunistic (Anypath) Routing

Opportunistic or anypath routing represents a significant advancement over traditional methods by leveraging the broadcast nature of wireless networks. Instead of forwarding packets to a single next-hop, a node broadcasts the packet to a set of candidate forwarders. Any node in this forwarding set that successfully receives the packet can forward it further toward the destination. This approach increases the probability of successful packet delivery, particularly in lossy and unpredictable environments. Protocols such as ExOR and MORE demonstrate the effectiveness of this strategy in improving throughput and reliability. However, most opportunistic routing schemes assume that packet losses across different links are independent, which is often not the case in real-world wireless networks.

1.3.4 Limitations of Existing Systems

Despite the advancements offered by these approaches, existing systems still face several important limitations. Many protocols ignore link rate diversity, treating all links as if they operate at the same transmission speed, which results in suboptimal routing decisions and inefficient use of network resources. Furthermore, the common assumption of independent link behavior does not reflect real-world conditions, where links often exhibit correlated losses due to shared interference, fading, or obstacles. This can lead to inaccurate estimation of delivery probabilities and inefficient forwarder selection. Consequently, these systems may generate redundant transmissions, increased congestion, and reduced overall performance, highlighting the need for more realistic and adaptive routing strategies.

Chapter 2

Literature Review

2.1 Introduction

The design of reliable multicast protocols for wireless multi-hop networks has been extensively studied to address the inherent challenges of lossy links, interference, and dynamic channel conditions. Traditional unicast-based approaches suffer from poor efficiency in one-to-many communication, while early multicast schemes often assume ideal lossless channels or independent link failures, which do not hold in real-world wireless environments. Recent advances leverage opportunistic routing (OR), network coding (NC), multi-rate adaptation, and forwarder cooperation to achieve higher throughput and reliability. These techniques exploit the broadcast nature of wireless medium, reduce redundant transmissions, and adapt to varying link qualities. However, most existing protocols fall short in simultaneously handling dynamic multi-rate systems, correlated link losses, multi-hop reliability, and real-time implementation feasibility—key requirements for practical deployment in mobile or time-varying scenarios such as vehicular networks, disaster recovery, or live content distribution. This literature review examines six representative works that form the foundation for the proposed reliable multi-hop multicast protocol. Each contribution is analyzed for its strengths and limitations with respect to the target problem: achieving end-to-end reliability in dynamic multi-rate wireless systems where link correlations exist and real-time performance is essential.

2.2 Multicast Least Cost Anypath Routing (M-LCAR)

This paper extends Least Cost Anypath Routing (LCAR) to multicast by computing ETX-based route costs to the destination set and all its subsets using a Bellman-Ford-style algorithm. It introduces semi-optimal and heuristic methods for forwarder-set selection and packet forwarding strategy, along with split-horizon to avoid loops. [Link to proposed topic: M-LCAR](#) directly addresses reliable multi-hop multicast with opportunistic forwarder sets and cost-based

routing, providing a foundation for subset-based cost calculation in multi-destination scenarios. However, it assumes single-rate operation and independent link losses, and does not consider dynamic environments or real-time implementation. The proposed protocol builds upon M-LCAR's anypath cost framework while extending it to multi-rate adaptation and correlated losses.

2.3 MORE: MAC-Independent Opportunistic Routing Encoding

MORE is the first practical protocol that combines random linear network coding with opportunistic routing. It eliminates the need for coordination by randomly mixing packets, runs directly over 802.11, and supports both unicast and multicast. Experimental results on a 20-node testbed show significant throughput gains over ExOR. [Link to proposed topic](#): MORE demonstrates the power of intra-flow NC for reliable multicast without scheduler overhead, making it highly relevant for real-time scenarios. It handles lossy links effectively but operates at a fixed single rate and assumes independent losses. The proposed work integrates MORE-style NC while adding multi-rate support and explicit correlation modeling for dynamic systems.

2.4 CodePipe: Reliable Multicast with Pipelined Network Coding

CodePipe introduces four innovations: LP-based opportunistic routing structure, opportunistic feeding (destinations act as cooperative sources once decoded), fast batch moving (pipelined batch dissemination), and inter-batch coding. It includes an online extension for dynamic node join/leave. Simulations show 2.56× throughput over MORE and better energy efficiency. [Link to proposed topic](#): This is one of the closest prior works, addressing reliable multi-hop multicast, dynamic networks, and real-time batch pipelining. It explicitly handles dynamic membership, which aligns with the proposed protocol's real-time requirements. However, CodePipe still assumes independent losses and single-rate links. The proposed design extends CodePipe's opportunistic feeding and pipelining to multi-rate environments with correlated link modeling.

2.5 Reliable Relay Assisted Wireless Multicast Using Network Coding

The authors propose a network-coding-based cooperative multicast scheme (NCBC) with one relay node. They analyze outage probability, diversity order, throughput, delay, and queue length for three protocols (direct, relay-assisted, and NC-based). The NC scheme reduces retransmissions by combining lost packets. [Link to proposed topic](#): It highlights the benefits of relay cooperation + NC for reliability in multi-hop-like settings and provides analytical

models for delay/queue length—critical for real-time performance. The single-relay focus is a simplification; the proposed protocol generalizes this to multi-hop, multi-rate, and correlated scenarios while retaining the NC-based retransmission efficiency.

2.6 Polynomial-Time Algorithms for Multirate Anypath Routing

This work presents the first optimal polynomial-time algorithms for multirate anypath routing. It jointly optimizes forwarding sets and transmission rates per node, generalizing Dijkstra/Bellman-Ford for multi-rate hypergraphs. The algorithms remain optimal even under correlated link losses. Trace-driven evaluation on an 802.11b testbed shows up to 6.4× improvement over single-rate anypath. [Link to proposed topic](#): This is the most directly relevant paper for the multi-rate aspect. It explicitly supports correlation between links (a core requirement of the proposed work) and provides efficient routing algorithms suitable for real-time computation. The proposed protocol adopts Laufer’s multirate anypath framework as the underlying routing layer while extending it to multicast reliability and dynamic adaptation.

2.7 Pacifier: High-Throughput Reliable Multicast

Pacifier integrates tree-based opportunistic routing, intra-flow NC, source rate limiting, and round-robin batching to solve the “crying baby” problem. It guarantees 100% packet delivery ratio while significantly improving throughput for both well-connected and poorly connected receivers. Testbed results show 83–144% gain over MORE. [Link to proposed topic](#): Pacifier excels in reliable multicast with fairness and high throughput in real-world lossy networks. Its round-robin batching supports real-time fairness, but it lacks multi-rate adaptation and link correlation handling. The proposed protocol incorporates Pacifier’s fairness and batching mechanisms into a multi-rate, correlation-aware design.

2.8 Research Gap and Motivation

While the reviewed works collectively advance opportunistic routing, network coding, multi-rate adaptation, and reliability, no single protocol jointly addresses dynamic multi-rate operation, correlated link losses, end-to-end multicast reliability, and real-time implementation constraints in multi-hop wireless systems. The proposed protocol fills this gap by integrating multirate anypath routing (with correlation support), pipelined NC with opportunistic feeding, and dynamic adaptation mechanisms, enabling practical real-time deployment.

Chapter 3

Proposed Work : Multirate Multicast Routing

The project presents a Correlation-Aware Multirate Multicast Routing approach for wireless multi-hop networks, which extends existing anypath routing concepts by incorporating link correlation along with link cost and transmission rate. The key objective is to improve reliability and efficiency by jointly considering realistic wireless network conditions that are often ignored in traditional models.

3.1 The Proposed Solution

The proposed protocol provides an enhanced and practical solution for multicast routing in dynamic environments:

- **Correlation-Aware Optimization:** The protocol incorporates inter-link correlation into routing decisions, recognizing that packet losses across links are not independent. By modeling shared interference and environmental effects, it improves the accuracy of delivery probability estimation and forwarder selection.
- **Joint Consideration of Link Cost and Rate:** Similar to multirate anypath routing, the proposed approach jointly optimizes link cost (ETX), link rate, and forwarding set. This ensures that both reliability and transmission efficiency are balanced, leading to reduced delay and improved throughput.
- **Efficient Forwarder Set Selection:** The protocol selects an optimal forwarding set of neighboring nodes for multicast transmission. By leveraging opportunistic routing, multiple nodes can forward packets, increasing the probability of successful delivery while minimizing redundant transmissions through correlation-aware filtering.

3.2 Proposed Method in the Context of Routing Algorithms

The proposed work can be viewed as an extension of existing anypath and multirate routing algorithms with added real-world considerations:

- **Adaptive and Scalable Design:** The protocol dynamically adapts to changes in link quality, transmission rates, and network conditions, making it suitable for large-scale and highly dynamic wireless networks.
- **Improved Efficiency:** By combining link cost, rate, and correlation, the approach reduces unnecessary retransmissions and improves bandwidth utilization compared to traditional and single-rate opportunistic routing methods.
- **Performance Benefits:** Simulation results demonstrate that the proposed method achieves higher packet delivery ratio, lower transmission overhead, and better overall network performance compared to existing systems that ignore link correlation and rate diversity.

3.3 Determining ETT

3.3.1 One Source Two Destination

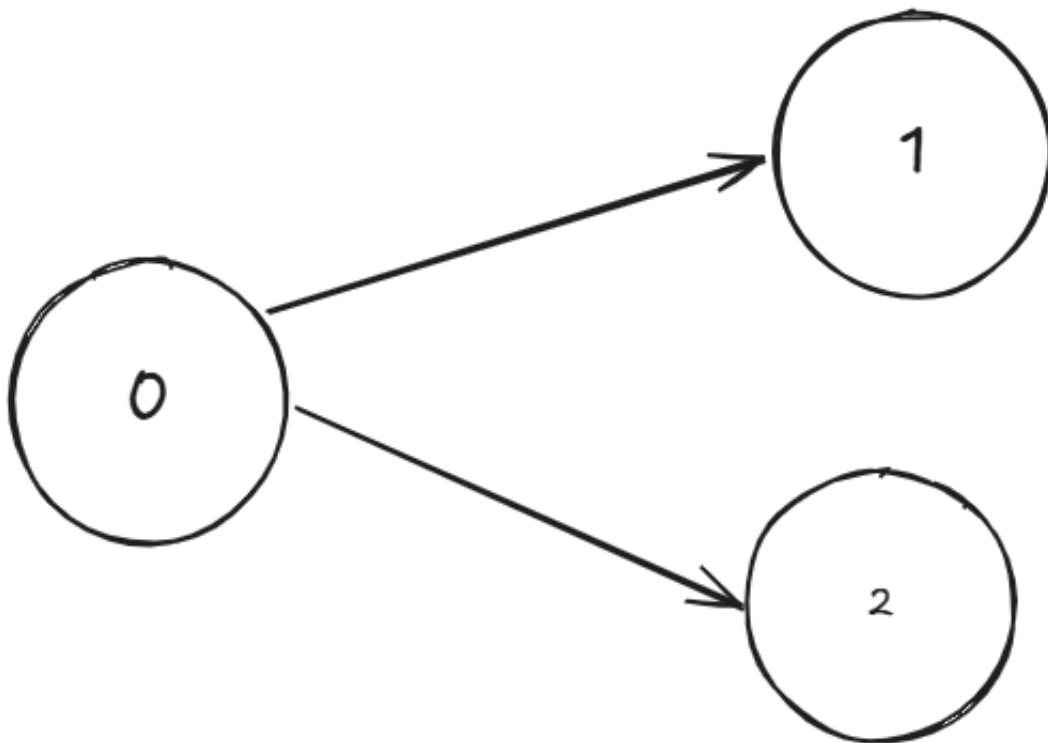


FIGURE 3.1: Single Source Two Destination

Input

- $ETX(0-1)=2$ $LinkRate(0-1)=1\text{mbps}$
- $ETX(0-2)=4$ $LinkRate(0-2)=3\text{mbps}$

Pseudocode**Algorithm 1** Compute Multicast ETX for Two Destinations from One Source

```

1: Transmit a packet from source node 0
2: if packet does not reach to any node then
3:    $x \leftarrow (1 - p_1)(1 - p_2)(1 + x)$ 
4: else if packet reaches to node 1 then
5:    $x \leftarrow p_1(1 - p_2)(1 + ETT_{0,2})$ 
6: else if packet reaches to node 2 then
7:    $x \leftarrow p_2(1 - p_1)(1 + ETT_{0,1})$ 
8: else
9:    $x \leftarrow p_1p_2(1 + \max(ETT_{0,1}, ETT_{0,2}))$ 
10: end if
11: Combine all cases:
12:  $x \leftarrow (1 - p_1)(1 - p_2)(1 + x) + p_1(1 - p_2)(1 + ETT_{0,2})$ 
13:    $+ p_2(1 - p_1)(1 + ETT_{0,1}) + p_1p_2(1 + \max(ETT_{0,1}, ETT_{0,2}))$ 
14: Solve for  $x$ 

```

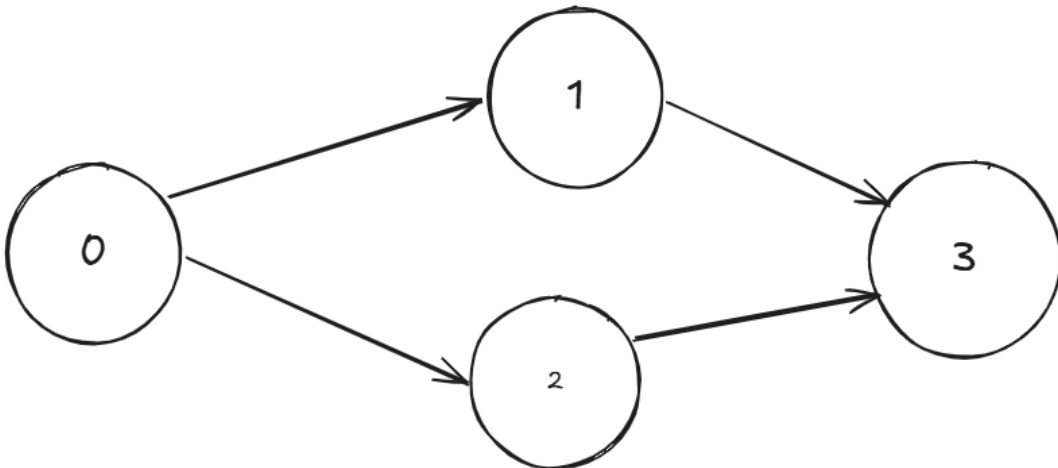
3.3.2 One source Two Intermediate Single Destination

FIGURE 3.2: Single Source Single Destination Using Two Intermediate

Input

- ETX(0-1)=2 LinkRate(0-1)=1mbps
- ETX(0-2)=4 LinkRate(0-2)=3mbps
- ETX(1-3)=3 LinkRate(1-3)=2mbps
- ETX(2-3)=1 LinkRate(2-3)=4mbps

ALGORITHM 2: Compute ETX and ETT for 1 Source, 1 Destination, 2 Intermediate Nodes

```

1: Compute  $q_1 \leftarrow 1/a$ ,  $q_2 \leftarrow 1/b$ 
   — Compute ETX —
2: if packet does not reach any node then
3:    $x \leftarrow (1 - q_1)(1 - q_2)(1 + x)$ 
4: else if packet reaches via path 2 only then
5:    $x \leftarrow (1 - q_1) q_2 (1 + d)$ 
6: else if packet reaches via path 1 only then
7:    $x \leftarrow q_1 (1 - q_2) (1 + c)$ 
8: else
9:    $x \leftarrow q_1 q_2 (1 + \min(c, d))$ 
10: end if
11: Combine all cases:
12:  $N_{ETX} \leftarrow (1 - q_1)(1 - q_2) + (1 - q_1) q_2 (1 + d)$ 
13:    $+ q_1(1 - q_2)(1 + c) + q_1 q_2 (1 + \min(c, d))$ 
14:  $D_{ETX} \leftarrow 1 - (1 - q_1)(1 - q_2)$ 
15: Solve for ETX  $\leftarrow N_{ETX}/D_{ETX}$ 

   — Compute ETT —
16:  $t \leftarrow 1/\min(al, bl)$ 
17: if packet does not reach any node then
18:    $x \leftarrow (1 - q_1)(1 - q_2) (t + x)$ 
19: else if packet reaches via path 2 only then
20:    $x \leftarrow (1 - q_1) q_2 (t + d/dl)$ 
21: else if packet reaches via path 1 only then
22:    $x \leftarrow q_1 (1 - q_2) (t + c/cl)$ 
23: else
24:    $x \leftarrow q_1 q_2 (t + \min(c/cl, d/dl))$ 
25: end if
26: Combine all cases:
27:  $N_{ETT} \leftarrow (1 - q_1)(1 - q_2) \cdot t + (1 - q_1) q_2 (t + d/dl)$ 
28:    $+ q_1(1 - q_2)(t + c/cl) + q_1 q_2 (t + \min(c/cl, d/dl))$ 

```

29: $D_{ETT} \leftarrow 1 - (1 - q_1)(1 - q_2)$
 30: Solve for $ETT \leftarrow N_{ETT}/D_{ETT}$

3.3.3 One source Two Intermediate Two Destination

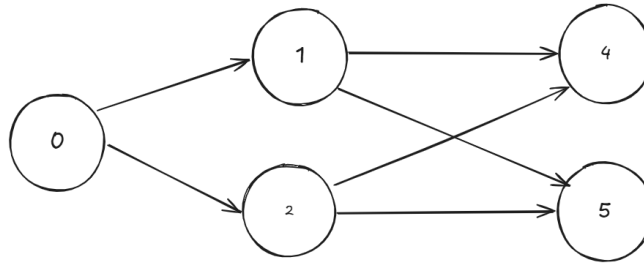


FIGURE 3.3: Single Source Two Destination Using Two Intermediate

Input

- $ETX(0-1)=2$ $LinkRate(0-1)=1\text{mbps}$
- $ETX(0-2)=4$ $LinkRate(0-2)=3\text{mbps}$
- $ETX(1-4)=3$ $LinkRate(1-4)=2\text{mbps}$
- $ETX(1-5)=1$ $LinkRate(1-5)=4\text{mbps}$
- $ETX(2-4)=2$ $LinkRate(2-4)=2\text{mbps}$
- $ETX(2-5)=7$ $LinkRate(2-5)=3\text{mbps}$

Forwarder Set and Destination Costs

	4	5	$\{4, 5\}$
2	$ETX(2-4)$	$ETX(2-5)$	$ETX(2-\{4,5\})$
1	$ETX(1-4)$	$ETX(1-5)$	$ETX(1-\{4,5\})$
$\{1, 2\}$	$\min(ETX(1-4), ETX(2-4))$	$\min(ETX(1-5), ETX(2-5))$	still have to write

TABLE 3.1: Forwarder Set vs Destination Set

ALGORITHM 3: Compute Multicast ETX: 1 Source (0), 2 Intermediate Nodes (1,2), 2 Destinations (3,4)

Require: $a = \text{ETX}(0, 1)$, $b = \text{ETX}(0, 2)$, $c = \text{ETX}(1, 5)$, $d = \text{ETX}(1, 4)$, $e = \text{ETX}(2, 5)$,
 $f = \text{ETX}(2, 4)$

Ensure: Multicast ETX from source 0 to destinations $\{5, 4\}$

— **Step 1: ETX from each intermediate to destinations** —

1: $\text{ETX}_{1 \rightarrow \{5,4\}} \leftarrow \text{ETX_2DEST1SRC}(c, d)$

2: $\text{ETX}_{2 \rightarrow \{5,4\}} \leftarrow \text{ETX_2DEST1SRC}(e, f)$

— **Step 2: Joint delivery probabilities to each destination** —

3: $q_{15} \leftarrow 1/c$, $q_{25} \leftarrow 1/e$

4: $q_{14} \leftarrow 1/d$, $q_{24} \leftarrow 1/f$

5: $p_5 \leftarrow q_{15} + q_{25} - q_{15} q_{25}$

▷ Prob. destination 5 receives

6: $p_4 \leftarrow q_{14} + q_{24} - q_{14} q_{24}$

▷ Prob. destination 4 receives

— **Step 3: ETX from source to destinations** —

7: $q_1 \leftarrow 1/a$, $q_2 \leftarrow 1/b$

8: **if** packet reaches neither intermediate **then**

9: contribution $\leftarrow (1 - q_1)(1 - q_2)$

10: **else if** packet reaches intermediate 1 only **then**

11: contribution $\leftarrow q_1 (1 - q_2) (1 + \text{ETX}_{1 \rightarrow \{5,4\}})$

12: **else if** packet reaches intermediate 2 only **then**

13: contribution $\leftarrow (1 - q_1) q_2 (1 + \text{ETX}_{2 \rightarrow \{5,4\}})$

14: **else**

15: best $\leftarrow \min(\text{ETX}_{1 \rightarrow \{5,4\}}, \text{ETX}_{2 \rightarrow \{5,4\}}, c + f, d + e)$

16: contribution $\leftarrow q_1 q_2 (1 + \text{best})$

17: **end if**

18: **Combine all cases:**

19: $N \leftarrow (1 - q_1)(1 - q_2) + q_1(1 - q_2)(1 + \text{ETX}_{1 \rightarrow \{5,4\}})$

20: $+ (1 - q_1) q_2 (1 + \text{ETX}_{2 \rightarrow \{5,4\}})$

21: $+ q_1 q_2 (1 + \min(\text{ETX}_{1 \rightarrow \{5,4\}}, \text{ETX}_{2 \rightarrow \{5,4\}}, c + f, d + e))$

22: $D \leftarrow q_1 + q_2 - q_1 q_2$

23: **return** N/D

3.3.4 One source Three Intermediate Two Destination

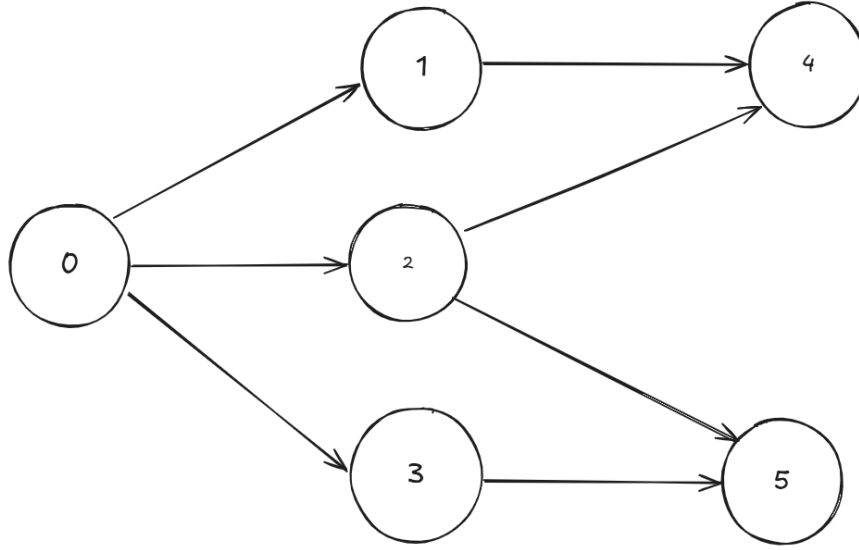


FIGURE 3.4: Single Source Two Destination Using Three Intermediate

Input

- $ETX(0-1)=2$ $LinkRate(0-1)=1\text{mbps}$
- $ETX(0-2)=4$ $LinkRate(0-2)=3\text{mbps}$
- $ETX(0-3)=1$ $LinkRate(0-3)=2\text{mbps}$
- $ETX(1-4)=3$ $LinkRate(1-4)=2\text{mbps}$
- $ETX(2-5)=1$ $LinkRate(2-5)=4\text{mbps}$
- $ETX(2-4)=2$ $LinkRate(2-4)=2\text{mbps}$
- $ETX(3-5)=7$ $LinkRate(3-5)=3\text{mbps}$

ALGORITHM 4: Find Optimal Forwarder Set: Source 0 to Destinations $\{4, 5\}$

Require: All link ETX and capacity values as above

Ensure: Minimum ETT and optimal forwarder set

— Candidate Forwarder Sets —

1: $F \leftarrow [\{2\}, \{1, 3\}, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}]$

— Precompute ETT for each candidate set —

2: // Forwarder set $\{2\}$

3: $x_1 \leftarrow ETX_2DEST1SRC(e, f, el, fl)$

4: $cost[0] \leftarrow b/bl + x_1$

5: // Forwarder set $\{1, 3\}$

6: $x_2 \leftarrow ETX_2DEST13(a, d, c, g, al, dl, cl, gl)$

```

7: cost[1]  $\leftarrow x_2$ 
8: // Forwarder set {1, 2}
9:  $x_4 \leftarrow \text{ETX\_2DEST12}(a, al, b, bl, d, dl, e, el, f, fl)$ 
10: cost[2]  $\leftarrow x_4$ 
11: // Forwarder set {2, 3}
12:  $x_6 \leftarrow \text{ETX\_2DEST23}(b, bl, c, cl, e, el, f, fl, g, gl)$ 
13: cost[3]  $\leftarrow x_6$ 
14: // Forwarder set {1, 2, 3}
15:  $x_3 \leftarrow \text{ETX\_1DEST12}(a, b, d, dl, e, el)$ 
16:  $x_5 \leftarrow \text{ETX\_1DEST23}(b, c, f, fl, g, gl)$ 
17:  $x_9 \leftarrow \text{ETX\_1DEST123\_TO45}(a, al, b, bl, c, cl, d, dl,$ 
18:  $e, el, f, fl, g, gl, x_1, x_3, x_5)$ 
19: cost[4]  $\leftarrow x_9$ 
    — Find Minimum Cost Forwarder Set —
20: minETT  $\leftarrow \text{cost}[0]$ , bestSet  $\leftarrow F[0]$ 
21: for  $i \leftarrow 1$  to 4 do
22:     if cost[ $i$ ] < minETT then
23:         minETT  $\leftarrow \text{cost}[i]$ 
24:         bestSet  $\leftarrow F[i]$ 
25:     end if
26: end for
27: Output minimum ETT = minETT
28: Output optimal forwarder set = bestSet

```

3.3.5 One source Three Intermediate Three Destination

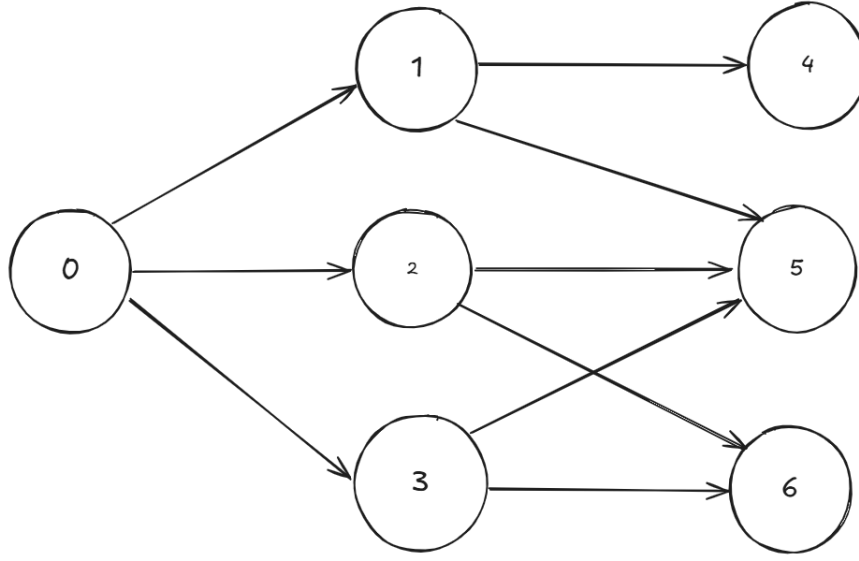


FIGURE 3.5: Single Source Two Destination Using Three Intermediate

Input

- ETX(0-1)=2 LinkRate(0-1)=1mbps
- ETX(0-2)=4 LinkRate(0-2)=3mbps
- ETX(0-3)=1 LinkRate(0-3)=2mbps
- ETX(1-4)=3 LinkRate(1-4)=2mbps
- ETX(1-5)=8 LinkRate(1-5)=3mbps
- ETX(2-5)=1 LinkRate(2-5)=4mbps
- ETX(2-6)=2 LinkRate(2-6)=2mbps
- ETX(3-5)=7 LinkRate(3-5)=3mbps
- ETX(3-6)=3 LinkRate(3-6)=2mbps

ALGORITHM 5: Find Optimal Forwarder Set: Source 0 to Destinations {4, 5, 6} using 3 Intermediates

Require: All link ETX and capacity values ($a \dots i, al \dots il$)

Ensure: Minimum ETT and optimal forwarder set

1: Initialize a 7×3 matrix cost with all zeros

— **Candidate Forwarder Set {1, 2} (Column 0)** —

2: $\text{cost}[0][0] \leftarrow b/bl + g/gl$

3: $\text{cost}[1][0] \leftarrow \text{ETX_1DEST12}(a, al, b, bl, e, el, f, fl)$

```

4: cost[2][0] ← ETX_2DEST12(a, al, b, bl, e, el, f, fl, g, gl)
5: cost[3][0] ← (a/al) + (d/dl)
6: cost[4][0] ← ETX_2DEST13(a, d, b, g, al, dl, bl, gl)
7: cost[5][0] ← ETX_2DEST12(b, bl, a, al, f, fl, e, el, d, dl)
8: cost[6][0] ← ETX456DESTFROM12(a, b, d, e, f, g, al, bl, dl, el, fl, gl)
  — Candidate Forwarder Set {1, 3} (Column 1) —
9: cost[0][1] ← c/cl + i/il
10: cost[1][1] ← ETX5DESTFROM123(a, b, c, e, f, h, al, bl, cl, el, fl, hl)
11: cost[2][1] ← ETX_2DEST12(a, al, c, cl, e, el, h, hl, i, il)
12: cost[3][1] ← (a/al) + (d/dl)
13: cost[4][1] ← ETX_2DEST13(a, d, c, i, al, dl, cl, il)
14: cost[5][1] ← ETX_2DEST12(c, cl, a, al, h, hl, e, el, d, dl)
15: cost[6][1] ← ETX456DESTFROM12(a, c, d, e, h, i, al, cl, dl, el, hl, il)
  — Candidate Forwarder Set {1, 2, 3} (Column 2) —
16: cost[0][2] ← ETX_1DEST12(b, bl, c, cl, g, gl, i, il)
17: x3 ← cost[0][2]
18: cost[1][2] ← ETX5DESTFROM123(a, b, c, e, f, h, al, bl, cl, el, fl, hl)
19: cost[2][2] ← ETX56DESTFROM123(a, b, c, e, f, g, h, i, al, bl, cl, el, fl, gl, hl, il, x3)
20: x4 ← cost[2][2]
21: cost[3][2] ← (a/al) + (d/dl)
22: cost[4][2] ← ETX46DESTFROM123(a, b, c, d, g, i, al, bl, cl, dl, gl, il, x3)
23: cost[5][2] ← ETX45DESTFROM123(a, b, c, d, e, f, h, al, bl, cl, dl, el, fl, hl)
24: cost[6][2] ← ETX456DESTFROM123(a, b, c, d, e, f, g, h, i, al, bl, cl, dl, el, fl, gl, hl, il, x4)
  — Find Minimum Cost Forwarder Set —
25: F ← [{1, 2}, {1, 3}, {1, 2, 3}]
26: ans_set ← F[0]
27: mini ← cost[6][0]
28: for j ← 1 to 2 do
29:   if cost[6][j] < mini then
30:     mini ← cost[6][j]
31:     ans_set ← F[j]
32:   end if
33: end for
34: Output "Minimum ETT from Node 0 to {4,5,6} is " + mini
35: Output "Path = " + ans_set

```

Chapter 4

Packet Simulation and Results

Packet simulation is a method used to analyze and evaluate the performance of network protocols in a controlled and virtual environment without requiring physical deployment. In wireless multi-hop networks, packet simulation helps model how data packets are generated, transmitted, forwarded, and received across nodes under different network conditions. It allows researchers to study the behavior of routing protocols by simulating real-world factors such as packet loss, interference, varying link quality, and dynamic transmission rates.

In packet-based simulation, the network is represented as a set of nodes connected through wireless links. Each packet generated at the source is transmitted through intermediate nodes (multi-hop) until it reaches one or more destination nodes in the case of multicast communication. The simulator tracks important events such as packet transmission, reception, collision, and loss. Based on these events, key performance metrics such as packet delivery ratio (PDR), end-to-end delay, throughput, and transmission overhead are calculated.

For this project, packet simulation plays a crucial role in evaluating the proposed multicast routing protocol. It helps analyze how effectively the protocol performs when incorporating link cost, link rate, and inter-link correlation. By simulating different network scenarios—such as varying node density, mobility, and channel conditions—the efficiency and reliability of the protocol can be tested and compared with existing approaches.

Overall, packet simulation provides a cost-effective and flexible platform to validate routing strategies before real-world implementation, ensuring that the protocol performs efficiently under diverse and dynamic wireless conditions.

4.1 Single Source Single Destination

Chapter 5

Conclusions and Future Work

5.1 Conclusion

This project presented a reliable multi-hop multicast routing approach for dynamic multi-rate wireless systems by incorporating link cost, link rate, and inter-link correlation into the routing process. Unlike traditional methods that assume independent links and fixed transmission rates, the proposed approach considers real-world wireless characteristics, leading to more accurate and efficient routing decisions. By leveraging opportunistic routing and correlation-aware forwarder selection, the protocol improves packet delivery reliability while reducing redundant transmissions.

The results demonstrate that integrating these factors significantly enhances overall network performance in terms of packet delivery ratio, transmission efficiency, and adaptability to changing network conditions. The proposed work highlights the importance of designing routing protocols that reflect practical wireless environments.

5.2 Future Work

While the proposed approach shows promising improvements, there are several directions for future enhancement. Future work will focus on evaluating the protocol in more complex network scenarios, including higher node densities, varying mobility patterns, and diverse traffic conditions to better reflect real-world deployments. This will help analyze the scalability and robustness of the protocol under different environments.

Additionally, the protocol can be extended to support energy-efficient routing for resource-constrained networks. Implementation in a real-world testbed can further validate its practical applicability. Further improvements may also include enhancing adaptability and optimizing performance under large-scale and highly dynamic wireless network conditions.

[1] [2] [3] [4] [5] [6]

Bibliography

- [1] S. P. Mazumdar, S. K. Bose, and W.-D. Zhong, “Multicast least cost anypath routing: Route cost calculations and forwarding,” *IEEE Communications Letters*, vol. 16, no. 10, pp. 1652–1655, 2012.
- [2] S. Chachulski, M. Jennings, S. Katti, and D. Katabi, “Trading structure for randomness in wireless opportunistic routing,” in *Proceedings of the 2007 ACM SIGCOMM Conference*, pp. 169–180, ACM, 2007.
- [3] P. Li, S. Guo, S. Yu, and A. V. Vasilakos, “Reliable multicast with pipelined network coding using opportunistic feeding and routing,” *IEEE Transactions on Parallel and Distributed Systems*, vol. 25, no. 12, pp. 3244–3256, 2014.
- [4] P. Fan, C. Zhi, C. Wei, and K. B. Letaief, “Reliable relay assisted wireless multicast using network coding,” *IEEE Journal on Selected Areas in Communications*, vol. 27, no. 5, pp. 749–762, 2009.
- [5] R. Laufer, H. Dubois-Ferrière, and L. Kleinrock, “Polynomial-time algorithms for multirate any-path routing in wireless multihop networks,” *IEEE/ACM Transactions on Networking*, vol. 20, no. 3, pp. 742–755, 2012.
- [6] D. Koutsonikolas, Y. C. Hu, and C.-C. Wang, “Pacifier: High-throughput, reliable multicast without “crying babies” in wireless mesh networks,” *IEEE/ACM Transactions on Networking*, vol. 20, no. 5, pp. 1375–1388, 2012.